Nirajan Paudel

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EDUCATION

University of Colorado Boulder — Boulder, CO
M.S. in Computer Science — Expected May 2027

Institute of Engineering, Tribhuvan University — Pokhara, Nepal
B.E. in Electronics, Communication & Information Engineering — 2020–2024

EXPERIENCE

Graduate Student Researcher, LECS Lab, University of Colorado Boulder — Oct 2025 – Present

- Co-authored *Speculative Decoding Across Languages* (arXiv:2605.30580; under review, ACL): evaluated speculative decoding on 11 languages and 2 model families (Qwen 3.5, Llama 3.2); showed draft/verifier divergence and acceptance rates track how well-resourced a language is.
- Built task-specific and general-domain distillation pipelines to reduce draft/verifier divergence; found distillation generalizes poorly across tasks for low-resource languages.
- Proposed n-gram draft models as a low-cost alternative to neural drafters, achieving higher speed-ups across all languages.

Machine Learning Intern, PalmMind Technology — Mar 2025 – May 2025

- Built production agentic RAG chatbots for EV dealers and insurance clients on GPT-4; migrated from LangChain to LangGraph for tool calling and parallel orchestrator-worker execution.
- Implemented LLM routing (GPT-4 for complex queries, GPT-3.5-turbo for classification) with prompt engineering and Redis caching, cutting API costs across 500+ customer inquiries.
- Improved retrieval accuracy for insurance-policy and vehicle-spec queries by integrating web scraping, Tesseract OCR, and context management.

Teaching Assistant, Pashchimanchal Campus, Tribhuvan University — Sep 2024 – Feb 2025

- Taught 90+ students C programming and Information Systems; mentored 15+ semester projects with 85% student proficiency in algorithm design.

PROJECTS

Tensor AI – Context-Aware Academic Assistant (RAG) — Nov 2025 – Present

- Built a production RAG pipeline (FastAPI, Pinecone, LlamaParse) over 10+ engineering programs, ingesting legacy PDFs with diagrams and tables.
- Added intent-based query routing and cross-encoder reranking (ms-marco-MiniLM-L-6-v2) with dynamic top-k (3–50); reranking improved Top-1 retrieval accuracy by 40% over plain vector search.
- Cut LLM API costs 70% with Redis caching of embeddings and classifications; achieved sub-200ms time-to-first-token via SSE streaming in a React frontend.

AttnLab – Mechanistic Interpretability Playground (attnlab.sarvabhaum.ai) — 2026

- Built and self-host a public web app (TransformerLens, PyTorch, FastAPI, React) showing per-head attention patterns from live forward passes on 5 models (GPT-2, Pythia, Qwen3), sent in a custom uint8 format instead of 151 MB of float32.
- Implemented a logit lens over every attention and MLP sub-layer and exact direct logit attribution to each head and MLP; recovers the IOI name-mover heads (L9H6, L9H9) unprompted and reproduces the reference notebook number for number.
- Built a tokenizer lab comparing 9 tokenizers (GPT-2, Llama 2, cl100k, o200k, XLM-R) on FLORES+ text in 32 languages, measuring each language's token premium over English (up to 16x for Burmese under GPT-2) and replaying real BPE merges.

Serverless vs. Kubernetes NLP Inference Benchmark — Oct 2025 – Dec 2025

- Deployed DistilBERT sentiment analysis on AWS Lambda and EKS with Terraform and GitHub Actions CI/CD; compared latency, cost, and scalability under load using Locust and Kubernetes HPA.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, Java, SQL, Bash
ML/AI: PyTorch, TensorFlow, HuggingFace, TransformerLens, LangChain, LangGraph, OpenAI/Anthropic APIs, RAG, Speculative Decoding, Mechanistic Interpretability, Transformers
Cloud & DevOps: AWS (Lambda, EKS, EC2, ECR), GCP (GKE, GCS), Kubernetes, Docker, Terraform, GitHub Actions, CI/CD
Tools & Databases: FastAPI, Flask, Redis, Pinecone, PostgreSQL, Git, Pandas, NumPy